Bypassing the Transport Plan: Dynamic Reweighting for Out-of-Distribution Detection with Optimal Transport

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Abstract

*Semi-supervised learning (SSL) has achieved remarkable progress by leveraging both limited labeled data and abundant unlabeled data. However, unlabeled datasets often contain out-of-distribution (OOD) samples from unknown classes, which can lead to performance degradation in open-set SSL scenarios. Current OOD detection methods are constrained by the absence of labeled OOD samples. Although optimal transport (OT) has proven to be effective to provide pseudo OOD scores for supervised learning, it still faces two main challenges, i.e., the unavoidable problem of finding the optimal transport plan and the unreliable OOD score caused by dense solutions. To overcome thess limitations, we propose a novel open-set OOD detection model named **DREW**, which leverages **D**ynamic **REW**eighting approach for OT-based OOD detection. Specifically, we start by formulating OOD detection as a semi-unbalanced optimal transport (SemiUOT) problem. The proposed DREW model can dynamically transform SemiUOT into the classical OT formula and then directly obtain the pseudo OOD score from the new source distribution weights. Contrary to existing OT-based OOD detection methods, the proposed DREW model provides theoretically grounded and more accurate pseudo OOD scores. Empirical results demonstrate the superiority of DREW in terms of both accuracy and efficiency.*

1. Introduction

Semi-supervised learning (SSL) is a pivotal machine learning paradigm that leverages abundant unlabeled data alongside limited labeled data [2, 9, 27, 40, 51, 57]. Traditional SSL relies on the assumption that both labeled and unlabeled data share the same class space and distribution [8, 11, 32, 50]. However, this closed-world setting is often unrealistic. In real-world open-set SSL scenarios, unlabeled data frequently contain out-of-distribution (OOD) samples from unknown classes, which can severely degrade model performance.

To alleviate this problem, researchers begin to explore the identification of outliers [24, 41, 45], namely OOD detection. While one possible solution is to directly implement OOD detection during testing [17, 18, 21, 34, 39], this approach only considers the testing phase with the model trained on purely in-distribution (ID) data. In open-set SSL, the model must be jointly optimized on both labeled and unlabeled data while ensuring the OOD detection module learns effectively without compromising classification accuracy. Consequently, [15, 53] propose to integrate OOD detection with semi-supervised learning during the training phase. However. the absence of reliable OOD labels hinders effective learning and the joint optimization framework risks sacrificing closed-set classification accuracy.

To address this challenge, recent works [42, 44] introduce third-party proxies into the training phase. For instance, [44] employs a one-vs-all (OVA) classifier for OOD detection. Optimal transport (OT), as an idea mathematical tool for the measurement of different distributions, has proven to be effective in finding OOD samples [25, 42].

However, existing OT based methods typically focus on finding the optimal transport plan for a given OT problem. The optimal transport plan gives the mass of each sample transferred from the source distribution to each sample in the target domain, and then are used determines the probability of whether a sample belongs to OOD based on the total mass

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of the transferred sample. Intuitively, it is unnecessary to find the complete solution of the transport plan in the above process. we are concerned with the sample-to-distribution matching relationship, but rather with the sample-to-sample matching relationship. Furthermore, in order to find the optimal transport plan, a common compromise is to obtain a dense approximation of the original problem through entropy regularization. The complete solution of the transport plan is redundant and unnecessary, which reduces the reliability of the obtained OOD score and thus leads to suboptimal performance of the learned model.

To resolve this dilemma, we model the OOD detection of each batch as a semi-unbalanced optimal transport (SemiUOT) problem and propose the dynamic reweighting approach, which dynamically converts the SemiUOT problem into classical OT and thus directly obtain the pseudo OOD score from the new source distribution weights, eliminating the need for entropy regularization or ad-hoc artificial weight amplification. This allows to obtain more accurate pseudo OOD scores for the training of the model. A novel open-set OOD detection model named **DREW** is proposed, which learns under the supervision of the **D**ynamic **REW**eighting module. In addition, a fast approximation algorithm with a theoretically analyzed error bound is proposed for optimization. Extensive experiments demonstrate the superiority of DREW over existing methods.

We summarize our main contributions as follows:

- We propose to dynamically convert the SemiUOT problem into classical OT to directly obtain reliable pseudo OOD score, bypassing the solution of transport plan and overcoming key limitations of existing OT based OOD detection methods.
- A novel open-set OOD detection framework DREW is constructed, which learns under the supervision of the OT module. For the optimization of DREW, we propose a fast approximation algorithm with a theoretically analyzed error bound.
- Extensive experiments demonstrate that DREW achieves competitive, and often superior, performance compared to state-of-the-art methods. Ablation and analytical studies provide insights into the properties and contributions of the proposed method.

2. Related works

2.1. Optimal transport

OT quantifies the discrepancy or distance between two distributions by calculating the minimum transport cost [7, 10, 26, 36–38]. As a mathematical tool, recent advances have shown the promising potential of OT for various machine learning tasks, such as natural language processing [3, 48], computer vision [4, 14, 23, 35], graph matching and representation [12, 55, 56], generative models [22, 31, 49], and reinforcement learning [1, 28, 47]. Classical OT is the most basic form of OT theory, where both marginal equality constraints on source and target distributions are preserved.

Definition 2.1 (Classical OT). *Given two distributions $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$, each containing M and N units,$\boldsymbol{a}$ and $\boldsymbol{b}$ are the weights vectors of them, and $\boldsymbol{C} \in \mathbb{R}_+^{M\times N}$ is the transport cost matrix. The classical OT is to find the optimal transport plan $\boldsymbol{\pi} \in \mathbb{R}_+^{M\times N}$ in the feasible solution set* Π.

$$\begin{aligned} &\min_{\boldsymbol{\pi}\in\Pi(\boldsymbol{\alpha},\boldsymbol{\beta})} \langle \mathbf{C}, \boldsymbol{\pi}\rangle_F = \sum_{i=1}^{M}\sum_{j=1}^{N} C_{i,j}\pi_{i,j}, \\ &\Pi(\boldsymbol{\alpha},\boldsymbol{\beta}) := \left\{\boldsymbol{\pi} \in \mathbb{R}_+^{M\times N} : \boldsymbol{\pi}\mathbf{1}_N = \boldsymbol{a}, \boldsymbol{\pi}^{\mathrm{T}}\mathbf{1}_M = \boldsymbol{b}\right\}. \end{aligned} \tag{1}$$

Since classical OT is essentially a linear optimization process, it can be solved exactly using any linear solver. However, when trying to make OT work in the real world, things are often trivial. If one of the two marginal constraints is relaxed, SemiUOT is defined as follows.

Definition 2.2 (SemiUOT with KL-Divergence). *SemiUOT is to find the optimal transport plan when the target constraint on the source distribution is removed [30], the formulation is,*

$$\begin{aligned} &\min_{\boldsymbol{\pi}\in\Pi(\boldsymbol{\alpha},\boldsymbol{\beta})} \langle \mathbf{C}, \boldsymbol{\pi}\rangle_F + \tau_a KL(\boldsymbol{\pi}\mathbf{1}_N \| \boldsymbol{a}), \\ &\Pi^s(\boldsymbol{\alpha},\boldsymbol{\beta}) := \left\{\boldsymbol{\pi} \in \mathbb{R}_+^{M\times N} : \boldsymbol{\pi}^{\mathrm{T}}\mathbf{1}_M = \boldsymbol{b}\right\}, \end{aligned} \tag{2}$$

where τ_a is the weight parameter. The divergence term KL is used to measure the discrepancy between the marginal distribution of the transport plan and the source distribution. A common approach to SemiUOT involves introducing an entropy regularization term and solving the approximated solution with a Sinkhorn [5]-like algorithm.

2.2. Out-of-distribution detection

Existing OOD detection methods can be categorized into test-time detection methods, neural networks based methods, and third-party proxy based methods.

Test-time detection. The test-time detection methods only deal with OOD samples at test phase [17, 18, 21, 34]. [18] using the probability distribution output by softmax layer to detect OOD samples. [34] aligns the energy score with the probability density of input by mapping it to a scalar energy value. GradNorm [21] performs detection by leveraging the vector norm of gradient extracted from pre-trained models.

Neural networks based methods. Methods based solely on neural networks focus on training neural networks to detect OOD samples [6, 13]. [53] proposes a joint optimization framework to perform OOD detection and semi-supervised learning, alternately updating relevant parameters and OOD score. DS3L [15] introduces a new weight function, selectively using unlabeled data and preventing performance degradation through a dual optimization mechanism. However, lack of reliable labels for supervision limits the OOD detection performance of these methods.

Third-party proxy based methods. For the lack of labels, methods begin to introduce third-party proxy for better detection. [44] employs a one-vs-all (OVA) classifier [43] for OOD detection. [25, 42] introduce OT for the OOD detection task, which obtain pseudo OOD score from the optimal transport plan obtained via the entropy regularized OT.

2.3. Optimal transport for OOD detection

The motivation for the OT-based OOD detection is intuitive. For the probability distribution of both the source and the target domain (e.g., the feature embedding of unlabeled and labeled data), the common assumption is that they both obey the uniform discrete distribution. If we model OOD detection problem as classical OT on this basis, it is difficult to distinguish ID samples from OOD samples via the transport plan $\boldsymbol{\pi}$. Thus, [42] enlarges the weights of the unlabeled data to force the marginal constraint of target domain (i.e., $\boldsymbol{\pi}^{\mathrm{T}}\mathbf{1}_N = \boldsymbol{a}$) to be relaxed. The probability that the sample with less total transmission quality is an OOD sample is greater, and vice versa. Then, the pseudo OOD score obtained via the OT module can be computed as, $\mathbf{S} = \frac{1}{k}\boldsymbol{\pi}\mathbf{1}_N$, assuming that the target domain is the feature embedding of unlabled data. Due to the mechanism above, it is unrealistic to deploy it directly during testing. A more practical approach is to use $\mathbf{S}$ as the labels to train the neural OOD detector. However, the redundant mass k based OOD detection lacks theoretical support and the solution of $\boldsymbol{\pi}$ via the common Sinkhorn-like OT solvers risk producing unreliable pseudo OOD score. This paper proposes to model the OOD detection task as SemiUOT and solve it by bypassing the optimal transport plan in a more accurate way.

3. Methodology

3.1. Problem Definition

Source and target domain sample: The source domain sample $\boldsymbol{X}_u \in \mathbb{R}^{M\times H\times W\times CH}$ is composed of unlabeled data with known classes and unlabeled data with unknown classes (i.e., OOD sample), where M is the batch size of the source domain sample and H,W, CH are the height, width and number of channels of the input image date respectively. The target domain sample is the labeled data $\boldsymbol{X}_l \in \mathbb{R}^{N\times H\times W\times CH}$ with known classes, where N is the batch size of the target domain sample. Both the labeled sample and the unlabeled sample with known classes are called ID sample. Note that M should be larger than N to comply with the setting of open-set SSL.

Open-set semi-supervised OOD detection: Given $\boldsymbol{X}_u$ and $\boldsymbol{X}_l$, the open-set semi-supervised OOD detection is to detect OOD samples in $\boldsymbol{X}_u$ with $\boldsymbol{X}_l$ as a reference and perform semi-supervised learning over both $\boldsymbol{X}_u$ and $\boldsymbol{X}_l$. This paper focuses on training neural networks to predict the OOD probability $\hat{\boldsymbol{S}} \in \mathbb{R}_+^{M+N}$ under the supervision of pseudo labels obtained by OT and preserving the closed-set classification accuracy through self-supervised regularization.

3.2. Overview

As shown in Figure 1, DREW is composed of two components, namely the semi-supervised learning module and the OOD detection module. The closed-set classification module g adopts the same architecture as [42, 46]. For a batch of sample from $\boldsymbol{X}_u$ and $\boldsymbol{X}_l$, we perform weak augmentation and strong augmentation on $\boldsymbol{X}_u$ to obtain $\boldsymbol{X}_w, \boldsymbol{X}_s \in \mathbb{R}^{M\times H\times W\times CH}$. The specific augmentation method is consistent with [46]. The augmented source and target sample are mapped to their feature embedding $\boldsymbol{Z} = f(\boldsymbol{X}) \in \mathbb{R}^d$ through feature encoder f, where d is dimension of feature embedding. The pseudo OOD scores $\boldsymbol{S}$ of the unlabeled samples are obtained from the OOD detection module, based on which the OOD loss is given:

$$\mathcal{L}_{ood} = \frac{1}{M+N}\|\hat{\boldsymbol{S}} - \boldsymbol{S}'\|_2^2, \tag{3}$$

where $\hat{\boldsymbol{S}} = h(\boldsymbol{X}_l \oplus \boldsymbol{X}_w)$ is obtained via the OOD detection head h. $\boldsymbol{S}' = \mathbf{1}_N \oplus \boldsymbol{S}$ and $\oplus$ indicates the concat operation. Putting together, the total loss is,

$$\mathcal{L} = \mathcal{L}_x + \mathcal{L}_u + \gamma\mathcal{L}_{ood}, \tag{4}$$

where $\mathcal{L}_x$ and $\mathcal{L}_u$ are supervised loss and unsupervised loss in Fixmatch [46] and γ denotes the balanced parameter. In the following section, we will further introduce the details of our proposed $\mathcal{L}_{ood}$ via dynamic reweighting approach.

3.3. OOD Detection Module for DREW

Although we can first solve for the complete optimal transport plan and then obtain the corresponding pseudo OOD score, in reality, in OOD detection tasks, we only care about the marginal distribution of the optimal transport plan, not the complete optimal transport plan. In this section, we propose the dynamic reweighting based OOD detection module, which incorporates a dynamic reweighting mechanism to estimate pseudo-OOD scores for unlabeled samples in a more accurate and computationally efficient manner, while being underpinned by rigorous theoretical guarantees. Firstly, We adopt the formulation of SemiUOT as Equation (2) to model the OOD detection task as the SemiOT problem, where $\boldsymbol{\alpha} = \sum_{i=1}^{M}\frac{1}{M}\delta_{\boldsymbol{u}_i}, \boldsymbol{\beta} = \sum_{i=1}^{N}\frac{1}{N}\delta_{\boldsymbol{l}_i}$ are the distribution weight of source samples and target samples respectively, and δ is the Dirac function. Given the feature embedding of labeled sample $\boldsymbol{Z}_l$ and that of strong augmented unlabeled sample $\boldsymbol{Z}_s$, we calculate the transport cost from $\boldsymbol{Z}_s$ to $\boldsymbol{Z}_l$ as,

$$\boldsymbol{C} = \mathbf{1}_{M\times N} - \frac{\boldsymbol{Z}_s\boldsymbol{Z}_l}{\|\boldsymbol{Z}_s\|_2^2\|\boldsymbol{Z}_l\|_2^2}. \tag{5}$$

Then, instead of directly solving the SemiUOT problem via Sinkhorn-like approach, we have the following proposition

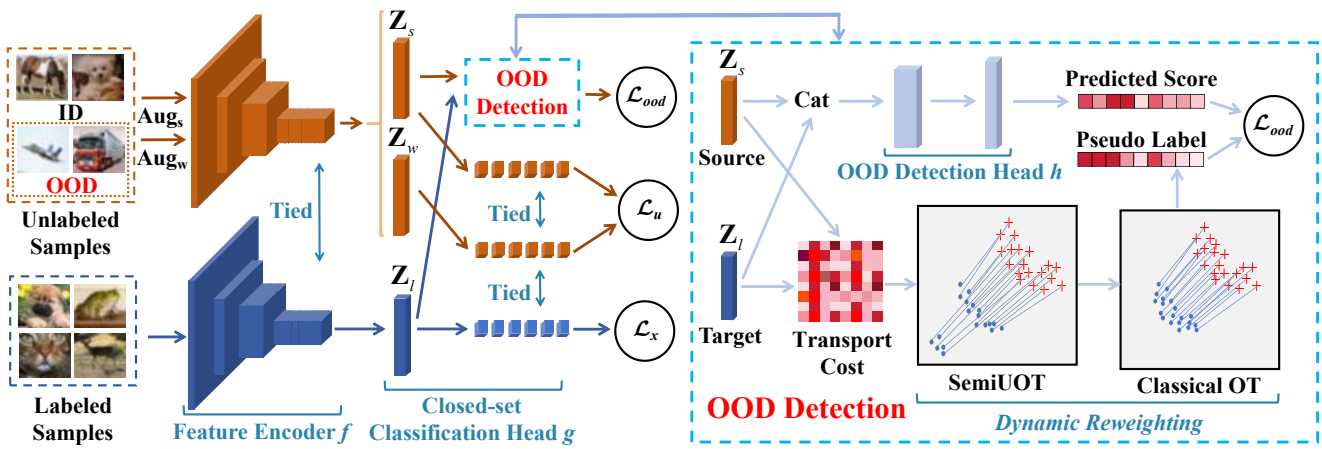

Figure 1. The left part of the figure depicts the overall framework of DREW. The right part of the figure shows the details of the dynamic reweighting based OOD detection module. "Tied" indicates weight sharing. Brown and dark blue arrows represent the flow of data in the target and source domains, respectively. The light blue arrows indicate the flow of data within the OOD detection module.

to transform the SemiUOT problem to classical OT and the problem can be tackled in a more accurate and efficient way.

Proposition 3.1 *(Dynamic Reweighting with Exact SemiUOT Solver). Given SemiUOT with KL-Divergence shown in Equation (2), we can rewrite its dual form as below:*

$$\begin{aligned} &\min_{\boldsymbol{u},\boldsymbol{v},\boldsymbol{s},\boldsymbol{\zeta}} \mathcal{J} = \tau_a \left\langle \boldsymbol{a}, \exp\left(-\frac{\boldsymbol{u}+\boldsymbol{\zeta}}{\tau_a}\right)\right\rangle - \left\langle \boldsymbol{v}-\boldsymbol{\zeta}, \boldsymbol{b}\right\rangle, \\ &s.t.\ u_i + v_j + s_{ij} = C_{ij}, s_{ij} \geq 0, \end{aligned} \tag{6}$$

where $\boldsymbol{u}$, $\boldsymbol{v}$, $\boldsymbol{s}$, and $\boldsymbol{\zeta}$ are dual variables. Equation (13) can be rewritten as classical OT:

$$\begin{aligned} &\min_{\boldsymbol{\pi}\geq 0} \mathcal{J}_{\mathrm{S}} = \langle \boldsymbol{C}, \boldsymbol{\pi}\rangle, \\ &s.t.\ \boldsymbol{\pi}\mathbf{1}_N = \boldsymbol{a} \odot \exp\left(-\frac{\boldsymbol{u}^* + \boldsymbol{\zeta}^*}{\tau_a}\right), \boldsymbol{\pi}^{\mathrm{T}}\mathbf{1}_M = \boldsymbol{b}, \end{aligned} \tag{7}$$

where $\boldsymbol{u}^*$ and $\boldsymbol{\zeta}^*$ are the optimal value of $\boldsymbol{u}$ and $\boldsymbol{\zeta}$.

Please refer to the appendix for detailed proof. That is, if we partially solve $\boldsymbol{u}$ and $\boldsymbol{\zeta}$, SemiUOT can be simplified to classical OT. Popular and efficient exact solvers for the subproblems in $\boldsymbol{u}$ and $\boldsymbol{\zeta}$ is L-BFGS [33]. The key insight is that solving the simplified classical OT for the OOD score using an exact OT solver is equivalent to obtaining it directly from the source weight in Equation (14).

The above approach is to obtain the equivalent form of SemiUOT via dynamically reweighting the weights of the source distribution. By eschewing the entropy regularization responsible for unreliable scores and the arbitrary amplification of source weights, DREW achieves more accurate OOD detection without the computational burden of directly solving the complete optimal transport plan.

3.4. Approximating SemiUOT for Optimization

Within the framework of Dynamic Reweighting, we can avoid solving for redundant optimal transport plans when calculating pseudo OOD scores. However, finding the exact solution is non-trivial. For the efficient optimization of DREW, we propose to approximating the formulation of SemiUOT and thus it can be tackled via the fixed point approach. To begin with, we review the exact SemiUOT:

$$\begin{aligned} &\min_{\boldsymbol{u},\boldsymbol{\zeta}} \mathcal{J}_{\mathrm{S}} = \tau_a \sum_{i=1}^{M} a_i \exp\left(-\frac{u_i+\boldsymbol{\zeta}}{\tau_a}\right) - \sum_{j=1}^{N} \mathcal{M}_j b_j, \\ &\text{where } \mathcal{M}_j = \left[\inf_{k\in[M]} [C_{kj} - u_k] - \boldsymbol{\zeta}\right]. \end{aligned} \tag{8}$$

Although $\mathcal{J}_{\mathrm{S}}$ is convex and has unique solutions, the presence of $\inf(\cdot)$ renders it a non-smooth function, leading to inefficient optimization. Thus, to accelerate the optimization process, we turn to the approximate solution with the following proposition.

Proposition 3.2 *(Accelerating dynamic reweighting with approximation). We consider a smooth approximation of SemiUOT to replace $\inf(\cdot)$ as $\inf_{k\in[M]}[C_{kj} - u_k] \approx -\epsilon \log[\sum_{k=1}^{M} e^{\frac{u_k - C_{kj}}{\epsilon}}]$. Note that $\epsilon > 0$ denotes the balanced hyperparameter among precision and smoothness of the function. Smaller ϵ (e.g., ϵ approaches to 0) could lead to more accurate while less smooth solutions. Then we can obtain the proposed Approximate SemiUOT Equation as $\widehat{\mathcal{J}}_{\mathrm{S}}$ by replacing $\inf(\cdot)$ with the smoothness term for $\widehat{u}$,*

$$\begin{aligned} \min_{\boldsymbol{u},\boldsymbol{\zeta}} \widehat{\mathcal{J}}_{\mathrm{S}} = \tau_a \exp\left(-\frac{\boldsymbol{\zeta}}{\tau_a}\right) \sum_{i=1}^{M} a_i \exp\left(-\frac{u_i}{\tau_a}\right) + \\ \sum_{j=1}^{N} \left[\epsilon \log\left[\sum_{k=1}^{M} \exp\left(\frac{u_k - C_{kj}}{\epsilon}\right)\right] + \boldsymbol{\zeta}\right] b_j. \end{aligned} \tag{9}$$

Then, u_i can be iteratively updated as follows:

$$u_i^{(l+1)} = \mathcal{T}(u_i^{(l)}) = \frac{\tau_a \epsilon}{\tau_a + \epsilon} \log\left(a_i \exp\left(-\frac{\boldsymbol{\zeta}}{\tau_a}\right)\right) - \frac{\tau_a \epsilon}{\tau_a + \epsilon} \log\left[\sum_{j=1}^{N}\left[\frac{\exp\left(-\frac{C_{ij}}{\epsilon}\right)}{\sum_{k=1}^{M}\exp\left(\frac{u_k^{(l)} - C_{kj}}{\epsilon}\right)}\right] b_j\right]. \tag{10}$$

Meanwhile, $\boldsymbol{\zeta}$ can eventually be computed as:

$$\boldsymbol{\zeta} = \tau_a \left[\log\left(\sum_{i=1}^{M} a_i \exp\left(-\frac{u_i}{\tau_a}\right)\right) - \log\left(\sum_{j=1}^{N} b_j\right)\right]. \tag{11}$$

With this approximation method, we can perform dynamic reweighting more efficiently. The proof is provided in the appendix. To evaluate the accuracy of the solutions obtained by the above approximation method, we analyze the strict error bound in the following proposition and find that the approximation error will become smaller with smaller ϵ.

Proposition 3.3 *(Approximation error). We consider the analysis between optimal results of $\boldsymbol{u}$ and $\widehat{\boldsymbol{u}}$ via setting $\boldsymbol{\zeta} = 0$ in SemiUOT. Then we define $E_p(\boldsymbol{u}) = \mathcal{J}_{\mathrm{S}}$ and $K_p(\widehat{\boldsymbol{u}}) = \widehat{\mathcal{J}}_{\mathrm{S}}$. Hence we have the following relationships: (1) $|K_P(\boldsymbol{u}) - E_P(\boldsymbol{u})| \leq \epsilon \log M$, (2) $|E_P(\widehat{\boldsymbol{u}}) - K_P(\widehat{\boldsymbol{u}})| \leq 0$. Thus we have the following error bound:*

$$|K_P(\boldsymbol{u}) - K_P(\widehat{\boldsymbol{u}})| \leq \epsilon \log M. \tag{12}$$

Therefore $\boldsymbol{u}$ and $\widehat{\boldsymbol{u}}$ will get closer with smaller ϵ.

The above proposition provides a strict upper bound on the error of the discrete approximation of the SemiUOT problem. That is, the error is proportional to $\epsilon \log M$, therefore the approximation accuracy can be controlled by adjusting the entropy regularization parameter ϵ, providing a theoretical guarantee for the numerical solution of the SemiUOT problem that the approximate solution and the true solution are almost identical in terms of the objective function value as long as ϵ is sufficiently small.

The detailed computation of the pseudo OOD scores based on the approximation term for dynamic reweighting is presented in Algorithm 1. The time complexity of this procedure is $O(MN/\eta_{\mathrm{err}})$, where η_{err} is the prescribed error tolerance such that $\widehat{\mathcal{J}}_{\mathrm{S}}(\boldsymbol{u}) \leq \widehat{\mathcal{J}}_{\mathrm{S}}(\boldsymbol{u}^*) + \eta_{\mathrm{err}}$. It is worth highlighting that the dynamic reweighting approach can directly obtain the OOD score from the source marginal distribution of $\widehat{\mathcal{J}}_{\mathrm{S}}$, thus avoiding the solution of $\boldsymbol{\pi}$. With the supervision of $\boldsymbol{S}'$, we train the OOD detection head h to distinguish the OOD samples from the ID ones. The detailed process for the training of DREW is provided in Algorithm 2.

Algorithm 1 Dynamic Reweighting for OOD Detection

Require: Optimization objective $\mathcal{J}_{\mathrm{S}}$.
Ensure: Pseudo OOD score $\boldsymbol{S}'$.
1: Initialize $t = 0$, $u^0 = (\frac{1}{M}, \frac{1}{M}, \cdots, \frac{1}{M})$ and $\boldsymbol{\zeta} = 0$.
2: **for** $t = 0, 1, 2, \cdots, T$ **do**
3: Obtain the optimal solution on $\boldsymbol{u}^{(t+1)}$ via the iteration according to Equation (16).
4: **end for**
5: Optimize $\boldsymbol{\zeta}$ by solving $\frac{\partial \mathcal{J}_{\mathrm{S}}}{\partial \boldsymbol{\zeta}} = 0$ via Equation (17).
6: Obtain the classical OT format $\widehat{\mathcal{J}}_{\mathrm{S}}$ via Equation (14).
7: The pseudo OOD score $\boldsymbol{S}$ of the unlabeled samples is computed as $\boldsymbol{S} = \boldsymbol{a} \odot \exp\left(-\frac{\boldsymbol{u}^* + \boldsymbol{\zeta}^*}{\tau_a}\right)$.
8: The pseudo OOD score of the labeled samples is $\mathbf{1}_N$.
9: **return** $\boldsymbol{S}' = \mathbf{1}_N \oplus \boldsymbol{S}$.

Algorithm 2 Open-Set Out-of-Distribution Detection

Require: Data loader, weak augmentation method A, strong augmentation method $\mathcal{A}$, encoder network $f(\cdot)$, closed-set classification head $g(\cdot)$ and OOD detection head $h(\cdot)$.
1: **for** $\boldsymbol{X}_l$,$\boldsymbol{X}_u$ in loader **do**
2: **Augmentation:** $\boldsymbol{X}_w, \boldsymbol{X}_s = A(\boldsymbol{X}_u), \mathcal{A}(\boldsymbol{X}_u)$.
3: **Encoding:** $\boldsymbol{Z}_l, \boldsymbol{Z}_w, \boldsymbol{Z}_s = f(\boldsymbol{X}_l), f(\boldsymbol{X}_w), f(\boldsymbol{X}_s)$.
4: **OOD detection:**
5: Compute cosine similarity $\boldsymbol{C}$ between $\boldsymbol{Z}_l$ and $\boldsymbol{Z}_s$.
6: Obtain OOD pseudo labels $\boldsymbol{S'}$ via the dynamic reweighting approach.
7: Obtain OOD predicted score $\hat{\boldsymbol{S}} = h(\boldsymbol{Z}_l, \boldsymbol{Z}_s)$.
8: Compute OOD loss $\mathcal{L}_{ood}$.
9: **Self-supervised classification:**
10: Obtain classification probability.
11: $\mathcal{Y}_l, \mathcal{Y}_w, \mathcal{Y}_s = g(\boldsymbol{Z}_l), g(\boldsymbol{Z}_w), g(\boldsymbol{Z}_s)$.
12: **Total loss:** $\mathcal{L} = \mathcal{L}_x + \mathcal{L}_u + \gamma \mathcal{L}_{ood}$.
13: Update the model via back propagation.
14: **end for**

4. Experiments

4.1. Setup

Datasets. We build open-set OOD benchmarks based on CIFAR-10 and CIFAR-100 [29] with different settings. By artificially dividing known classes as labeled samples, the models need to detect OOD samples exist in $\mathbf{X}_u$. Following the setup in [46] and [42], the complete training set is used as unlabeled data, while labeled data is randomly extracted from the training set and the number of samples for each known class is set to different values corresponding to different classification difficulties. We also evaluate DREW on the more challenging ImageNet-30 [19] dataset, where 20 classes are selected as the known classes, with the remaining as the unknown classes.

Table 1. Top-1 accuracy and AUROC tests were performed on the CIFAR-10 benchmark dataset under different settings. We tested the performance of different methods with 6 labeled classes and 50, 100, and 400 labeled samples per class. To verify the statistical significance of the results, we report the mean and standard deviation of 5 experiments. The best results for each setting are shown in bold.

# of Labeled Classes	50		100		400	
Metric	Acc	AUROC	Acc	AUROC	Acc	AUROC
MTCF	79.7±0.9	96.6±0.5	86.3±0.9	98.2±0.3	91.0± 0.5	98.9±0.1
T2T	88.2±0.7	75.5±0.5	89.0±1.0	77.2±0.2	90.3±0.5	82.3±0.2
OpenMatch	89.6±0.9	99.3±0.3	92.9±0.5	**99.7**±0.2	94.1±0.5	99.3±0.2
POT	92.1±0.2	**99.7**±0.1	92.9±0.2	99.5±0.1	93.6±0.1	99.4±0.1
DREW	**92.2**±0.3	99.6±0.1	**93.0**±0.3	99.5±0.2	**94.1**±0.2	**99.4**±0.1

Table 2. Top-1 accuracy and AUROC tests were performed on the CIFAR-100 benchmark dataset under different settings. We tested the performance of different methods when the number of labeled classes was 55 and 80, and the number of labeled samples per class was 50 and 100, respectively. We report the mean and standard deviation of 5 experiments. The best results for each setting are shown in bold.

# of Known	55				80			
# of Labeled	50		100		50		100	
Metric	Acc	AUROC	Acc	AUROC	Acc	AUROC	Acc	AUROC
MTCF	66.5±1.2	81.2±3.4	72.1±0.5	80.7±4.6	59.9±0.8	79.4±2.5	66.4±0.3	73.2±3.5
T2T	72.2±1.4	60.4±1.6	73.1±0.8	59.8±1.4	63.5±1.2	55.0±1.8	66.8±0.7	55.4±1.5
OpenMatch	72.3±0.4	87.0±1.1	75.9±0.6	86.5±2.1	66.6±0.2	86.2±0.6	70.5±0.3	86.8±1.4
POT	78.7±0.1	88.4±0.1	81.1±0.1	89.5±0.3	75.4±0.1	88.1±0.3	**78.1**±0.1	88.0±0.1
DREW	**78.8**± 0.2	**91.7**±0.3	**81.2**±0.4	**91.2**±0.4	**75.5**±0.3	**90.8**±0.1	78.0±0.3	**90.9**±0.2

Baselines. To demonstrate the superiority of our method on OOD detection benchmark, we compare our proposed DREW with comprehensive baselines, including Fixmatch[46], MTCF[52], T2T[20], OpenMatch[44], and POT[42]. Since the semi-supervised classification module of DREW in this paper adopts the Fixmatch structure, we consider it as a baseline. The maximum softmax prediction probability as the score function is used to give Fixmatch the ability to perform OOD detection during testing. MTCF and T2T are typical purely neural networks based methods. OpenMatch adopts a method based on OVA classifiers. In addition, to prove the effectiveness of DREW, we also compare DREW with other OT-based method.

Metrics. To measure the OOD detection results, we calculate the area under the receiver operating characteristic curve (AUROC) to observe the ability to identify OOD samples. In this paper, we focus on image classification tasks in the presence of OOD samples, so we also report closed-set classification accuracy. In scenarios where we focus on comparing the performance of different methods, we report top-1 accuracy and AUROC, and in scenarios where we focus on analyzing the change with different architectures or parameter settings, we report top-1 to top-5 accuracy to facilitate comprehensive analysis.

Implementation details. To be fair, we adopt Wide ResNet (WRN) [54] as the backbone, with WRN-28-2 on CIFAR-10 and WRN-28-8 on CIFAR-100, consistent with the baseline. Since Fixmatch is used in the closed-set semi-supervised classification module of DREW. The model is trained with a Nesterov SGD optimizer with 0.9 momentum and 5×10^{-4} weight decay. We implement the cosine annealing learning rate adjustment strategy and set the initial learning rate as 0.03. τ_a in Equation (2) is set to 0.01 when comparing SDM with baselines. Putting all the components together, we set the balance factor of L_x and L_u as 1 while setting that of L_{ood} as 0.01. For the ImageNet-30 dataset, We use ResNet-18 [16] as the backbone, the experimental settings follow the case on CIFAR-10/100. More settings and implementation details are provided in the appendix.

4.2. Main Results

The experimental results on CIFAR-10, CIFAR-100, and ImageNet-30 are summarized in Table 1, Table 2, and Table 3. Our analysis reveals a clear performance hierarchy. Methods relying solely on neural networks perform poorly across all datasets, lagging significantly behind proxy-based approaches. While OpenMatch shows competitive results on CIFAR-10, it falls short of OT-based methods, particularly on CIFAR-100 where its closed-set accuracy is lower. To more directly demonstrate the advantages of the proposed method, we also report the H-scores of the DREW model and the runner-up method in Figure 2 (d). Overall, DREW achieves better performance than POT and maintains competitive performance on the more challenging ImageNet-30 dataset with 224×224 resolution. The most significant advantage of DREW is observed in AUROC on CIFAR-100, a gain potentially attributable to the larger number of classes, which causes other methods to not be able to perform OOD

Table 3. Top-1 accuracy and AUROC on ImageNet-30 benchmarks. 20 classes are selected as known classes, and the remaining classes are unknown classes. We report the mean and standard deviation of 5 experiments. The best results for each setting are shown in bold.

Method	FixMatch	MTCF	T2T	OpenMatch	POT	Ours
Top1Acc	91.7±0.5	86.4±0.7	87.8±0.9	89.6±1.0	92.0±0.3	**92.1**±0.5
AUROC	45.1±1.2	93.8±0.8	55.7±10.8	96.4±0.7	97.4±0.4	**97.6**±0.1

Table 4. Ablation study on CIFAR-10 benchmark. To analyze the effectiveness of dyanmic reweighting, Fixmatch-OOD is aopted as the baseline, which uses the maximum softmax prediction probability to give Fixmatch the ability to perform OOD detection during training phase. To compare the improvement compared to other OT-based methods, we also present the results of Fixmatch-POT in the table.

# of Known	55				80			
# of Labeled	50		100		50		100	
Metric	Acc	AUROC	Acc	AUROC	Acc	AUROC	Acc	AUROC
Fixmatch-OOD	78.2±0.7	57.3±1.1	80.8±0.6	56.7±1.2	75.3±0.6	48.7±0.9	78.1±0.5	47.3±0.7
FixMatch+POT	78.7±0.1	88.4±0.1	81.1±0.1	89.5±0.3	75.4±0.1	88.1±0.3	**78.1**±0.1	88.0±0.1
Fixmatch+DREW	**78.8**±0.2	**91.7**±0.3	**81.2**±0.4	**91.2**±0.4	**75.5**±0.3	**90.8**±0.1	78.0±0.3	**90.9**±0.2

detection well.

4.3. Adaptation to different OT solvers

Since DREW dynamically converts the SemiUOT from $\boldsymbol{\alpha}$ to $\boldsymbol{\beta}$ in each batch into classical OT during training, in theory any classical OT solver can be connected in series behind DREW. In order to verify the adaptability of DREW to different OT solvers, we insert several OT solvers [5] into DREW for trial as follows:

- **EMD.** The optimal trasnsport plan between two distributions can be solved directly and acccurately by EMD without adding a regularization term.
- **Sinkhorn.** Sinkhorn incorporates entropy regularization into the optimal transport problem and uses iterative scaling to efficiently approximate the solution is a common approach for solving the OT problem. But the resulting approximately dense solution affects the model accuracy.
- **Entropic partial wasserstein (EPW).** EPW also introduces an entropy regularization term, but allows for distribution quality mismatch and is computationally efficient.

Besides, we also report the results of POT as a baseline in Table 5. The DREW plugged into three different OT solvers all outperformed POT in terms of metrics. Interestingly, when we directly connect modules similar to those in POT in series to DREW, we can also observe performance improvements. Since EMD obtains an exact solution, while others obtain approximate solutions, the OOD labels obtained based on DREW and EMD are more reliable.

4.4. Analysis and ablation study

1) Ablation study. Table 4 reports the overall performance of Fixmath-OOD, Fixmatch+POT and Fixmatch+DREW. Fixmatch itself cannot detect OOD samples accurately enough, even when performing OOD detection based on predicted probabilities during inference. When training OOD detector under the supervision of DREW, the AUROC metric which indicates the open-ser OOD detection quality reach an impressively higher lever compared to Fixmatch-OOD. Performance improvement of DREW does not come at the cost of testing latency because only Fixmatch and neural networks that output ID probability are activated during the training phase. Compared to POT, DREW brings improvements not only in OOD detection accuracy but also in classification accuracy.

2) Quantitative analysis of the pseudo labels. In Table 6, We compare the AUROC of the pseudo lables produced by DREW to that of POT with the additional parameter set to 1.25 and 2.5 respectively. All the experiments are conducted on test datasets with labeled training samples as the target samples. The pseudo labels produced by DREW are more accurate than POT. It is worth noting that since k is used to artificially amplify the weights of source domain samples, its determination is linked to the proportion of OOD samples in the source domain data. DREW, however, does not require such adjustments and can directly obtain the OOD score by bypassing the solution of the optimal transport plan.

3) Influence of $\mathcal{L}_{ood}$. Since DREW is a multi-task learning framework, the coefficients of the loss functions corresponding to different tasks are important. Figure 2 (a) depicts the changes of top-1 classification accuracy and OOD detection quality with γ corresponding to $\mathcal{L}_{ood}$. Besides, when the value of γ is too small, such as 0.01, the impact of $\mathcal{L}_{ood}$ on the overall loss is too small to make the quality of OOD detection worse. In other cases, DREW is stable enough to the changes of relevant core parameters.

4) Influence of τ_a. As we can see that, the larger value of τ_a in Equation (2) involves more data to get matched. Figure 2 (b) reports the results with τ_a changes to check the influence of this property on DREW. Given that τ_a is essentially a scaling factor, our experiments span a fairly

Table 5. Adaptation to different OT solvers. We report top-1 to top-5 accuracy and AUROC on CIFAR-100 with 55 known classes and 50 labeled samples per known class, with POT as the baseline. We compare the performance when the DREW approach are combined with EPW, Shinkhorn and EMD. The basic property of each OT solver is presented for comparison. The best results are shown in bold.

Metirc	Solver	Type	Solution	Acc					AUROC
				Top1	Top2	Top3	Top4	Top5	
POT	POT	Partial OT	Approximate	78.4	87.7	91.5	93.8	95.0	90.1
DREW(+EPW)	EPW	OT	Approximate	79.0	**88.1**	91.8	93.7	95.0	91.0
DREW(+Sinkhorn)	Sinkhorn	OT	Approximate	79.2	88.0	**91.9**	**93.8**	**95.3**	91.2
DREW(+EMD)	EMD	OT	Exact	**79.5**	88.0	91.8	93.7	95.2	**91.5**

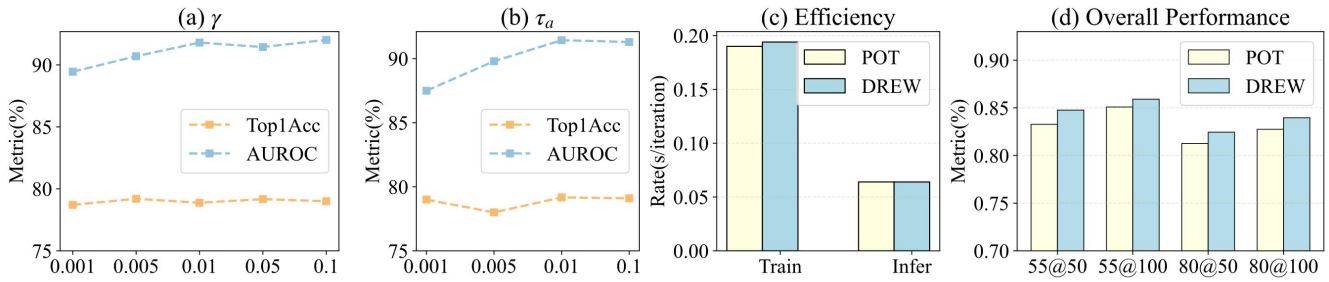

Figure 2. Analysis study. (a) The change of performance with γ set as 0.001, 0.005, 0.01 and 0.1. (b) The change of performance with τ_a set as 0.001, 0.005, 0.01, 0.05 and 0.1. (c) The efficiency comparison of P.(POT) and D.(DREW),where the train and infer rate (s/iteration) are reported. (d) The HS score comparison of POT and DREW. All results are on CIFAR-100 benchmark and the results of (a), (b), and (c) are with 55 known classes and 50 labeled samples per known class.

Table 6. Quantitative analysis of the accuracy of the pseudo labels on CIFAR-100. We report the performance of the OT module of both DREW and POT. We set the hyperparameter k of POT as 1.25 and 5 respectively to analyze its effect on OOD score quality.

# of Known	55		80	
# of Labeled	50	100	50	100
POT($k = 1.25$)	54.9	54.9	58.0	55.2
POT($k = 2.5$)	59.2	58.7	60.0	57.9
DREW	**59.7**	**59.2**	**60.1**	**58.3**

large range from 0.001 to 1, where no significant increase or decrease in performance is observed. Thus τ_a does not need to be overly fine-tuned to adapt to specific tasks and shows sufficient robustness.

5) Efficiency Analysis. To validate that the improvement of performance does not come at the cost of significant delay, we count the execution rate of DREW compared with the baseline. The training rate and testing rate are calculated as the average rate within 1024 iterations in Figure 2 (c). Owing to the acceleration approach of Proposition 3.2, the training latency of DREW is modest compared to POT. Consistent with the analysis in this paper, DREW cause no test delay. In addition, the training rate of DREW is 1.2s/iteration on the ImageNet-30 dataset, which means only 6× latency at 49× resolution compared to the case on CIFAR-10/100.

5. Conclusion

In summary, we propose a novel open-set semi-supervised OOD detection framework DREW, which aims at mitigating the core challenges of current OT-based OOD detection methods, i.e., the unavoidable problem of finding the optimal transport plan and the unreliable OOD score caused by dense solutions. Through theoretical derivation, we innovatively propose the dynamic reweighting approach that dynamically converts SemiUOT into classical OT and provides reliable pseudo OOD lables for supervision. The core insight is that the proposed approach can bypass the redundant solution of transport plan. The approximate SemiUOT equation is proposed for the efficient optimization of dynamic reweighting, with the analysis about the approximation error bound. Based on dynamic reweighting, DREW comprehensively surpasses all baselines on all benchmarks. Our proposal will motivate future work to pursue more efficient open-set learning paradigms.

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